

Article

Temporally Consistent Student Behavior Recognition in Smart Classrooms via Attention-Guided Perception and State Estimation

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Abstract

Recognizing student behaviors in classroom videos remains challenging due to complex backgrounds, frequent occlusions, subtle inter-class motion differences, and temporal jitter in frame-wise predictions. To address these issues, this paper proposes a hybrid student behavior recognition framework that integrates a Multi-branch Spatiotemporal Attention Network (MSTA-Net) with a Behavior State Kalman Filter (BSKF). At the perceptual level, MSTA-Net employs decoupled channel, spatial, and short-term temporal attention branches to enhance discriminative behavioral features while suppressing irrelevant background information. At the cognitive level, BSKF reformulates behavior recognition as a continuous state estimation problem in a high-dimensional probability space, where behavioral inertia is exploited to smooth noisy observations and improve temporal consistency. Experimental results on the SCB-Dataset and real-world classroom video sequences demonstrate that the proposed method achieves an accuracy of 94.7% and a real-time inference speed of 33 FPS. Compared with purely deep learning-based models, the proposed framework reduces the Action Category Switching (ACS) rate by 50%, indicating substantially improved robustness in long-term behavior recognition. These results suggest that coupling attention-based perception with Kalman-based state estimation provides an effective and efficient solution for reliable student behavior analysis in intelligent classroom environments.

Keywords: student behavior recognition; classroom video analysis; spatiotemporal attention; Kalman filtering; temporal consistency; smart education



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1. Introduction

Driven by the rapid development of artificial intelligence and national initiatives for AI-empowered education, smart classrooms have become an important direction in educational modernization [1]. Student behavior in classrooms provides valuable evidence for understanding learning states, engagement levels, and teaching effectiveness [2,3]. Traditional observation-based approaches, however, are often subjective, inefficient, and difficult to scale. In this context, computer vision-based automatic behavior recognition has attracted increasing research attention. Accurate student behavior recognition can provide teachers with real-time feedback for instructional adjustment, support administrators in objective teaching evaluation, and offer students personalized learning analytics for self-regulated learning.

Despite these advantages, robust student behavior recognition in real classroom environments remains challenging. Classroom scenes usually involve complex backgrounds, high student density, varying illumination, and frequent occlusions. In addition, many classroom behaviors are fine-grained and visually similar, such as reading, writing, listening, and using a mobile phone. These factors make it difficult for recognition models to consistently focus on discriminative students, body regions, and behavioral cues. Recent classroom behavior recognition studies have reported that crowded classroom environments, complex backgrounds, and severe occlusions can lead to feature confusion and degraded recognition performance [4,5].

Existing classroom behavior recognition methods still face several limitations. First, classroom videos contain a large amount of irrelevant visual information, which may distract the model from key behavioral regions and cause information overload. Second, frame-wise or short-clip-based prediction strategies often produce temporal jitter, resulting in frequent and unreasonable switching between behavior categories. Such abrupt transitions are inconsistent with the continuous nature of human behavior [6]. Third, conventional recurrent models may suffer from vanishing gradients and limited long-range temporal modeling capability, while recent Transformer-based video models have shown stronger potential for long-range spatiotemporal representation [7,8]. Finally, there remains a trade-off between recognition accuracy and real-time efficiency, as high-capacity models often require substantial computational resources and are therefore difficult to deploy in practical classroom scenarios.

These challenges can be understood from two complementary perspectives. At the perceptual level, the model needs to selectively extract discriminative features from cluttered visual inputs. Attention mechanisms are effective for enhancing salient regions and suppressing redundant information within frames or short video clips. Representative attention modules, such as SENet and CBAM, have demonstrated their effectiveness in channel recalibration and spatial localization [9,10]. However, attention-based models alone are generally insufficient to ensure temporal consistency across long video sequences. At the cognitive level, recognition results should evolve smoothly over time, reflecting the inertia and continuity of human behavior. Kalman filtering provides a principled framework for recursive state estimation and temporal smoothing [11], and has been explored for smoothing noisy behavior or gesture recognition outputs [12]. Nevertheless, its performance depends heavily on the reliability of the input observations. Therefore, attention mechanisms and Kalman filtering are naturally complementary: the attention network can generate high-quality behavior probability observations, while the Kalman filter can refine these observations through state prediction and uncertainty-aware correction.

Motivated by this observation, this paper proposes a hybrid student behavior recognition framework that integrates a Multi-branch Spatiotemporal Attention Network (MSTA-Net) with a Behavior State Kalman Filter (BSKF). Specifically, MSTA-Net serves as the perceptual observation module and decouples attention modeling into channel, spatial, and short-term temporal branches to capture discriminative behavioral features under complex classroom conditions. BSKF serves as the cognitive state estimation module and reformulates discrete behavior recognition as a continuous state estimation problem in a high-dimensional probability space. By exploiting behavioral inertia and dynamically adjusting observation noise according to prediction uncertainty, the proposed framework improves both recognition accuracy and temporal robustness.

The main contributions of this paper are summarized as follows.

- We propose a hybrid observation–estimation framework for classroom student behavior recognition, which systematically integrates attention-based visual perception with Kalman-based state estimation.

- We design MSTA-Net, a multi-branch spatiotemporal attention network that jointly models channel semantics, spatial saliency, and short-term temporal dependencies for fine-grained classroom behavior representation.
- We introduce BSKF, which maps discrete behavior categories into continuous probability state vectors and enables Kalman filtering to be applied to behavior recognition and temporal smoothing.
- We conduct comprehensive experiments, including ablation studies, comparative evaluations, and visualization analyses, to verify the effectiveness, real-time performance, and temporal stability of the proposed method.

The remainder of this paper is organized as follows. Section 2 reviews related work on classroom behavior recognition, attention mechanisms, and temporal smoothing. Section 3 presents the proposed MSTA-Net and BSKF framework. Section 4 describes the experimental setup and discusses the results. Section 5 concludes the paper and outlines future research directions.

2. Related Work

2.1. Deep Learning-Based Classroom Behavior Recognition

Early studies on human behavior and classroom activity recognition mainly relied on hand-crafted visual descriptors, such as Histogram of Oriented Gradients (HOG), Histogram of Optical Flow (HOF), and Spatiotemporal Interest Points (STIPs), followed by conventional classifiers such as support vector machines [13,14]. These methods are computationally efficient and easy to implement, but their representation capacity is limited. As a result, they are often sensitive to illumination changes, viewpoint variations, background clutter, and occlusions, which are common in real classroom environments.

With the development of deep learning, convolutional neural networks have become the dominant paradigm for video-based behavior recognition. Two-Stream Networks extract spatial appearance features from RGB frames and temporal motion features from optical flow, thereby improving action recognition performance by explicitly modeling complementary visual cues [15]. Subsequently, 3D convolutional architectures, including C3D, I3D, and SlowFast, further enhanced spatiotemporal feature learning by jointly modeling appearance and motion information within video clips [16–18]. These methods have achieved strong performance on general action recognition benchmarks.

However, directly applying general video recognition models to classroom scenarios remains challenging. Classroom videos usually contain dense student distributions, complex backgrounds, frequent mutual occlusions, and subtle inter-class behavioral differences. Recent classroom behavior recognition studies have shown that such factors may cause feature confusion and reduce detection reliability in real teaching environments [4,5]. To alleviate these issues, some studies have explored skeleton- or pose-based representations, which can reduce background interference and emphasize body-structure information [19]. In addition, temporal Transformer-based models have been introduced to enhance long-range dependency modeling in videos [7]. Nevertheless, pose estimation may become unreliable under severe occlusion, while large Transformer-based models often involve high computational costs. Therefore, achieving accurate, robust, and real-time student behavior recognition in complex classroom environments remains an open problem.

2.2. Attention Mechanisms and Spatiotemporal Feature Enhancement

Attention mechanisms have been widely adopted in computer vision to enhance discriminative features and suppress irrelevant information. In the channel dimension, SENet adaptively recalibrates feature responses by modeling inter-channel dependencies, allowing the network to emphasize informative semantic channels [9]. In the spatial

dimension, Spatial Transformer Networks learn geometric transformations to improve spatial alignment, while CBAM sequentially applies channel and spatial attention to refine feature maps and localize salient regions [10,20]. These attention modules provide effective mechanisms for improving feature selectivity in complex visual scenes.

In recent years, self-attention and Transformer architectures have achieved significant progress in image and video understanding. Vision Transformer (ViT) demonstrates that global self-attention can provide a powerful alternative to convolutional inductive biases in visual representation learning [21]. For video understanding, ViViT and TimeSformer extend self-attention to the spatiotemporal domain, enabling long-range interactions across frames [7,22]. Video Swin Transformer further improves efficiency through hierarchical representations and shifted window attention [8]. These methods have shown strong potential for capturing global temporal dependencies and complex motion patterns.

Despite these advances, most existing attention-based methods are not specifically designed for fine-grained classroom behavior recognition. Some methods focus mainly on channel recalibration or spatial localization, while others rely on global self-attention with considerable computational overhead. In classroom scenarios, discriminative behavioral cues are often distributed across multiple dimensions: channel-level semantic responses indicate what behavior is occurring, spatial attention identifies where the behavior-related regions are located, and temporal attention captures how micro-actions evolve over short time intervals. A recent survey on multimodal human action recognition also indicates that effectively combining CNN-based local feature extraction with attention-based global modeling remains an important research direction [23]. Therefore, a lightweight multi-branch attention mechanism that jointly considers channel semantics, spatial saliency, and temporal continuity is highly desirable for classroom behavior recognition.

2.3. Temporal Consistency and Kalman Filtering

Temporal consistency is a crucial requirement for video-based behavior recognition. Many existing deep learning models perform recognition on individual frames or short clips, and the final predictions are often generated independently over time. Although this strategy can achieve high static classification accuracy, it may produce unstable predictions in long video sequences. In particular, small visual disturbances, occlusions, or ambiguous postures may cause abrupt category changes between adjacent frames, resulting in temporal jitter. Non-local neural networks and other temporal modeling methods have attempted to strengthen long-range dependency modeling in video recognition [6], but prediction stability remains difficult to guarantee when visual observations are noisy or incomplete.

Kalman filtering provides a classical and principled framework for recursive state estimation under uncertainty [11]. It has been widely used in tracking and navigation tasks because of its ability to combine dynamic prediction with noisy observations. In visual tracking, SORT applies Kalman filtering to online object tracking, while DeepSORT further incorporates appearance embeddings to improve association robustness [24,25]. More recently, OC-SORT has improved motion-based association by rethinking observation-centric tracking strategies [26]. These studies demonstrate that Kalman filtering can effectively improve temporal stability when the target state can be represented in a continuous state space.

However, applying Kalman filtering to behavior recognition is not straightforward. Unlike object tracking, where states such as position and velocity are naturally continuous, behavior recognition usually produces discrete semantic labels or categorical probability distributions. Therefore, discrete behavior categories cannot be directly incorporated into the standard Kalman observation model without an appropriate state-space formulation. Consequently, existing temporal smoothing strategies often resort to heuristic post-processing methods, such as Median Filtering, Majority Voting, Exponential Mov-

ing Average (EMA), or Temporal Ensembling. While computationally lightweight, these heuristic approaches rely on fixed temporal windows or static decay factors, making them agnostic to the instantaneous confidence variations of the visual network. Alternatively, classical probabilistic graphical models, such as Hidden Markov Models (HMMs) and Conditional Random Fields (CRFs), have been utilized for structural post-processing. However, HMMs typically operate on discrete hidden states and struggle to seamlessly integrate with the high-dimensional continuous probability spaces of modern deep networks. Furthermore, CRFs often require computationally expensive iterative offline inference, inherently violating the strict real-time constraints required for smart classroom deployments. These severe limitations highlight the absence of an explicit, dynamic, and efficient probabilistic state estimation mechanism. Although Kalman filter-based smoothing has been explored in video-based gesture recognition [12], the systematic integration of attention-based behavior observation and Kalman-based state estimation remains insufficiently studied in classroom behavior recognition. This motivates the proposed framework, which maps behavior predictions into a continuous probability state space and exploits behavioral inertia to improve temporal robustness.

3. Methodology

This section presents the proposed student behavior recognition framework based on a coupled observation–estimation mechanism. The framework is designed to address two key challenges in classroom behavior recognition: discriminative feature extraction under complex visual conditions and temporal instability in long video sequences. To this end, a two-layer architecture is constructed. The lower layer is a Multi-branch Spatiotemporal Attention Network (MSTA-Net), which extracts discriminative behavioral features from classroom video clips and generates category-level observation probabilities. The upper layer is a Behavior State Kalman Filter (BSKF), which performs recursive state estimation based on the temporal evolution of behavior probabilities. Through this design, the recognition task is transformed from independent frame-wise or clip-wise classification into a continuous state estimation problem with temporal constraints.

3.1. Overall Architecture

As illustrated in Figure 1, the proposed framework follows a perception–cognition decoupling design. The framework transforms the student behavior recognition task from a traditional independent frame-wise classification into a continuous state estimation process with temporal constraints. Formally, let $\mathbf{X}$ be the input video clip and $\hat{y}_t$ be the final predicted behavior category at time t . The entire process can be summarized by the following operator-based representation:

$$\begin{cases} \mathbf{z}_t = \mathcal{A}_{msta}(\phi(\mathbf{X}); \omega_{a,b,c}) \\ \hat{\mathbf{x}}_t = \mathcal{K}_{bskf}(\mathbf{z}_t, \hat{\mathbf{x}}_{t-1}, E_t) \\ \hat{y}_t = \arg \max(\Pi(\hat{\mathbf{x}}_t^{1:N})) \end{cases} \quad (1)$$

where

- $\mathcal{A}_{msta}(\cdot)$ denotes the MSTa-Net operator, which maps the raw visual input to a posterior probability distribution $\mathbf{z}_t$ through a 3D backbone ϕ and multi-branch attention weights ω .
- $\mathcal{K}_{bskf}(\cdot)$ represents the Behavior State Kalman Filter operator, which recursively refines the state estimate $\hat{\mathbf{x}}_t$ by incorporating behavioral inertia and the adaptive uncertainty E_t .
- $\Pi(\cdot)$ is the projection operator that ensures the estimated state satisfies the constraints of the probability simplex.

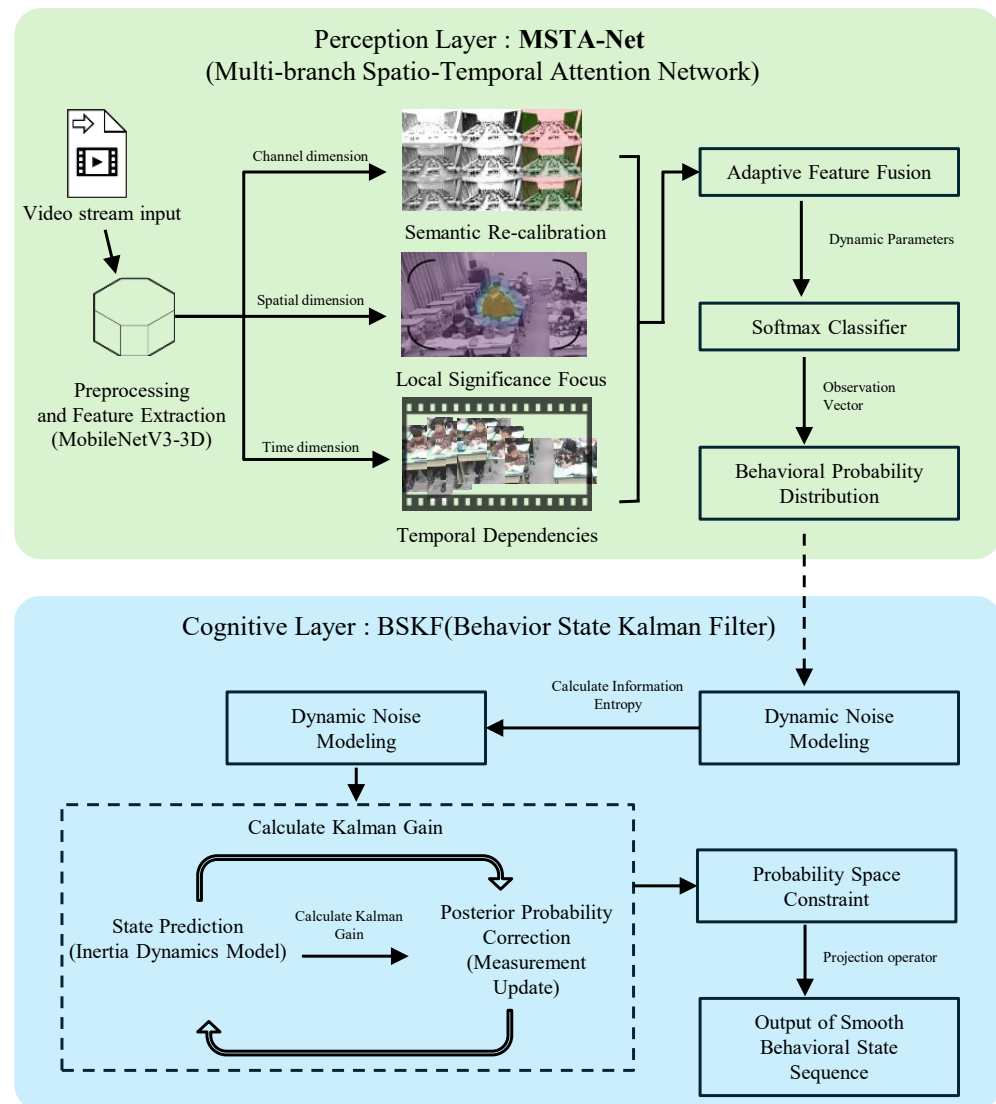


Figure 1. Dual-layer architecture for behavior recognition based on MSTA-Net and BSKF.

The perception layer is implemented by MSTA-Net, which simulates the bottom-up selective attention mechanism of human vision. It adaptively enhances behavior-related features along the channel, spatial, and short-term temporal dimensions. The cognition layer is implemented by BSKF, which simulates top-down temporal reasoning. By treating the output of MSTA-Net as a noisy observation, BSKF exploits behavioral inertia to produce a temporally consistent estimate $\hat{x}_t$. This closed-loop pipeline forms a robust observation–estimation system for complex classroom environments.

3.2. MSTA-Net: Multi-Branch Spatiotemporal Attention Observation Network

To balance recognition accuracy and real-time efficiency, MobileNetV3-3D [27] is adopted as the lightweight backbone of the perception layer. Compared with general action recognition datasets, the SCB-Dataset is specifically designed for classroom scenarios and contains a large number of fine-grained student behaviors with subtle visual differences [28,29]. Therefore, the proposed MSTA-Net is trained end-to-end on this dataset to learn robust and discriminative spatiotemporal representations under complex classroom conditions. The output of MSTA-Net is not only used for category prediction, but also serves as the observation input for the subsequent BSKF module.

3.2.1. 3D Backbone and Feature Encoding

The input video stream is first divided into short clips with a fixed temporal length of $T = 16$. Let the input clip be denoted as

$$\mathbf{X} \in \mathbb{R}^{3 \times T \times H \times W}, \quad (2)$$

where 3 denotes the RGB channels, and H and W represent the height and width of each frame, respectively. The 3D backbone maps the input clip into a high-level spatiotemporal feature tensor:

$$\mathbf{F} = \phi(\mathbf{X}), \quad \mathbf{F} \in \mathbb{R}^{C \times T' \times H' \times W'}, \quad (3)$$

where $\phi(\cdot)$ denotes the MobileNetV3-3D feature extractor, C is the number of feature channels, and T' , H' , and W' denote the temporal and spatial resolutions of the encoded feature map. Compared with 2D convolutional networks, 3D convolutions jointly encode appearance and local motion information, which is beneficial for distinguishing fine-grained classroom behaviors with subtle temporal variations.

3.2.2. Orthogonal Spatiotemporal Attention Decoupling

Classroom video features contain redundant information from background objects, neighboring students, and irrelevant body regions. To improve feature selectivity, MSTANet decouples attention modeling into three orthogonal branches: channel attention, spatial attention, and short-term temporal attention. These branches, respectively, answer three complementary questions: what semantic features are important, where the behavior-related regions are located, and when the micro-action evolves.

Channel Semantic Recalibration. The channel attention branch focuses on behavior-related semantic responses. Inspired by channel recalibration mechanisms [9], global average pooling is applied to aggregate the spatiotemporal context of each channel. A lightweight two-layer multilayer perceptron is then used to model inter-channel dependencies and generate channel-wise attention weights:

$$\mathbf{F}'_c = \mathbf{F} \otimes \sigma(\text{MLP}(\text{GAP}(\mathbf{F}))), \quad (4)$$

where $\text{GAP}(\cdot)$ denotes global average pooling, $\sigma(\cdot)$ is the sigmoid activation function, and $\otimes$ represents element-wise multiplication. This branch suppresses background-related channels and enhances semantic channels associated with student behaviors.

Spatial Saliency Localization. The spatial attention branch aims to locate behavior-related regions. In classroom scenarios, many actions, such as writing, reading, and using a mobile phone, are strongly associated with local regions including the head, hands, and desk area. To capture both local and global spatial cues, local convolutional features and global context features are fused to generate a spatial attention map:

$$\mathbf{F}'_s = \mathbf{F} \otimes \sigma\left(\text{Conv}_{1 \times 1}\left([\mathbf{F}_{\text{local}}; \mathbf{F}_{\text{global}}]\right)\right), \quad (5)$$

where $[\cdot; \cdot]$ denotes feature concatenation, and $\text{Conv}_{1 \times 1}(\cdot)$ is used to learn the spatial attention projection. This branch guides the network to focus on discriminative body regions while reducing interference from classroom background clutter.

Short-term Temporal Dependency Modeling. The temporal attention branch captures short-term motion evolution across consecutive frames. The feature tensor is reshaped into a temporal sequence and processed by a lightweight Transformer encoder. Self-attention is used to model inter-frame dependencies and identify subtle temporal changes in micro-actions:

$$\mathbf{F}'_t = \text{LayerNorm}(\mathbf{F}_t + \text{MultiHeadAttention}(\mathbf{F}_t)), \quad (6)$$

where $\mathbf{F}_t$ denotes the temporally reshaped feature sequence. This branch enhances the representation of short-term behavior dynamics, which is particularly important for distinguishing visually similar classroom actions.

3.2.3. Adaptive Feature Fusion

After obtaining the outputs of the three attention branches, MSTA-Net adaptively fuses them using learnable weights. Let ω_a , ω_b , and ω_c denote the contributions of the channel, spatial, and temporal attention branches, respectively. To ensure stable fusion, the weights are normalized by a Softmax operation:

$$\omega_a + \omega_b + \omega_c = 1, \quad \omega_a, \omega_b, \omega_c \geq 0. \quad (7)$$

The fused feature representation is then computed as

$$\mathbf{F}_{fusion} = \omega_a \mathbf{F}'_c + \omega_b \mathbf{F}'_s + \omega_c \mathbf{F}'_t. \quad (8)$$

Finally, the fused representation is fed into a fully connected classifier followed by a Softmax function to generate the behavior observation vector:

$$\mathbf{z}_t = \text{Softmax}\left(H(\mathbf{F}_{fusion})\right), \quad (9)$$

where $H(\cdot)$ denotes the fully connected classification head. For the six-class behavior recognition task, $\mathbf{z}_t \in \mathbb{R}^6$ represents the posterior probability distribution over behavior categories at time t , satisfying

$$\sum_{i=1}^6 z_{t,i} = 1. \quad (10)$$

Specifically, these six core fine-grained student classroom behavior categories include: Listening (C_1), Writing (C_2), Reading (C_3), Using Phone (C_4), Sleeping (C_5), and Raising Hand (C_6). The observation vector $\mathbf{z}_t$ provides a probabilistic description of the current visual evidence and is subsequently used as the input observation for BSKF.

3.3. Behavior State Kalman Filter

Although MSTA-Net can produce discriminative behavior probability observations, its frame-wise or clip-wise outputs may still fluctuate due to occlusion, illumination changes, motion blur, and ambiguous postures. Such fluctuations can cause frequent and non-physical switching between behavior categories in long video sequences. To address this problem, we introduce a Behavior State Kalman Filter (BSKF), which reformulates student behavior recognition as a continuous state estimation problem in a probability space. Instead of directly using the raw output of MSTA-Net as the final prediction, BSKF treats it as a noisy observation and recursively estimates the latent behavior state by combining visual observations with temporal behavioral inertia.

3.3.1. Probability–Velocity State Space

For a behavior recognition task with N categories, the probability vector at time t is denoted as

$$\mathbf{p}_t = [p_{t,1}, p_{t,2}, \dots, p_{t,N}]^T, \quad (11)$$

where $p_{t,i}$ represents the confidence score of the i -th behavior category. In this paper, $N = 6$. To model not only the current behavior confidence but also its temporal variation, the behavior state is defined as a probability–velocity vector:

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{p}_t \\ \mathbf{v}_t \end{bmatrix} = [p_{t,1}, \dots, p_{t,N}, \dot{p}_{t,1}, \dots, \dot{p}_{t,N}]^T, \quad (12)$$

where $\mathbf{v}_t = \dot{\mathbf{p}}_t$ denotes the temporal change rate of the behavior probability vector. For the six-class recognition task, $\mathbf{x}_t \in \mathbb{R}^{12}$. This formulation enables the filter to track both the instantaneous behavior tendency and its evolving momentum, thereby providing a continuous representation for originally discrete behavior categories.

The observation vector generated by MSTA-Net is denoted as $\mathbf{z}_t \in \mathbb{R}^N$. Since MSTA-Net observes only the behavior probability rather than its velocity, the observation model is defined as

$$\mathbf{z}_t = \mathbf{H}\mathbf{x}_t + \mathbf{r}_t, \quad (13)$$

where $\mathbf{r}_t$ is the observation noise, and $\mathbf{H}$ is the observation matrix:

$$\mathbf{H} = \begin{bmatrix} \mathbf{I}_N & \mathbf{0}_N \end{bmatrix}. \quad (14)$$

Here, $\mathbf{I}_N$ is an $N \times N$ identity matrix, and $\mathbf{0}_N$ is an $N \times N$ zero matrix.

3.3.2. Behavioral Inertia Dynamics

Human classroom behaviors generally evolve continuously over time. For example, a student usually does not switch instantaneously and repeatedly between listening and writing within adjacent frames. Based on this observation, we employ a constant-velocity dynamic model to describe the evolution of behavior probabilities:

$$\mathbf{x}_t = \mathbf{F}\mathbf{x}_{t-1} + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad (15)$$

where $\mathbf{F}$ is the state transition matrix, $\mathbf{w}_t$ is the process noise, and $\mathbf{Q}$ is the process noise covariance matrix. The transition matrix is defined as

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_N & \Delta t \cdot \mathbf{I}_N \\ \mathbf{0}_N & \mathbf{I}_N \end{bmatrix}, \quad (16)$$

where Δt denotes the time interval between two consecutive observations.

The process noise covariance matrix controls the strength of behavioral inertia. It is defined as

$$\mathbf{Q} = \begin{bmatrix} \sigma_p^2 \mathbf{I}_N & \mathbf{0}_N \\ \mathbf{0}_N & \sigma_v^2 \mathbf{I}_N \end{bmatrix}, \quad (17)$$

where σ_p^2 controls the uncertainty of probability evolution, and σ_v^2 controls the uncertainty of probability velocity. A smaller σ_p^2 imposes stronger constraints on abrupt probability changes, while σ_v^2 allows moderate variations in behavioral tendency. In this work, σ_p^2 is set to a small value to suppress sudden non-physical probability jumps, while σ_v^2 is used to preserve the ability to respond to real behavior transitions.

The design of the coefficients within the state transition matrix $\mathbf{A}$ in Equation (16) and the process noise covariance matrix $\mathbf{Q}$ in Equation (17) directly governs the temporal evolutionary physics of the latent behavioral probability space. Specifically, the elements $A_{1,2} = \Delta t$ and $A_{2,2} = 1$ formalize a first-order kinematic propagation, under which the student's current behavioral state profile is assumed to drift smoothly based on its immediate historical evolutionary velocity.

In Matrix (17), the stochastic perturbations of the behavioral propagation are explicitly quantified by the coefficients parameterized by the state positioning variance σ_p^2 and the velocity volatility variance σ_v^2 . The analytical expansion of Q integrates these white noise variances over the continuous temporal increment Δt , yielding terms such as $\frac{1}{3}\Delta t^3\sigma_v^2$ and $\frac{1}{2}\Delta t^2\sigma_v^2$. These analytical coefficients function as a dynamic tolerance mechanism: a higher σ_v^2 mathematically accommodates rapid, spontaneous intentional transitions (e.g., suddenly transitioning from *Listening* to *Raising Hand*), preventing the filter from exhibiting excessive sluggishness. Conversely, the measurement noise scale coefficient ϵ reflects the instantaneous uncertainty of the vision-based front-end (MSTA-Net) caused by occlusions or volatile illumination in classroom environments. To ensure reproducibility and absolute transparency of the framework's configuration, all critical architectural and filtering hyperparameters are consolidated in Table 1.

Table 1. System configuration and hyperparameter specifications of the proposed framework.

Component	Symbol	Description	Default Value	Optimization Strategy
MSTA-Net	Λ	Attention regularization scaling	0.45	Cross-validation grid search
	τ	Temporal segment window length	16 frames	Empirical standard trade-off
BSKF Layer	Δt	Temporal sampling increment	0.04 s	Video frame rate matching (25 fps)
	σ_p^2	Latent state process noise variance	1.0×10^{-4}	Kinetic calibration
	σ_v^2	Evolutionary state velocity variance	2.5×10^{-3}	Kinematic calibration
	ϵ	Measurement variance regularizer	1.5×10^{-2}	Front-end error estimation

3.3.3. Entropy-Based Adaptive Observation Noise

In standard Kalman filtering, the observation noise covariance is usually fixed. However, the reliability of MSTANet observations may vary significantly across time in real classroom videos. For example, occlusion, poor illumination, motion blur, or visually similar behaviors may increase prediction uncertainty. Therefore, using a fixed observation noise covariance may be insufficient for robust temporal smoothing.

To address this issue, BSKF introduces an entropy-based adaptive observation noise strategy. Given the behavior probability observation $\mathbf{z}_t$, its information entropy is computed as

$$E_t = - \sum_{i=1}^N z_{t,i} \log(z_{t,i} + \epsilon), \quad (18)$$

where ϵ is a small constant used to avoid numerical instability. A low entropy value indicates that the network prediction is concentrated on one behavior category and therefore more reliable. In contrast, a high entropy value indicates that the prediction distribution is ambiguous and the observation should be trusted less.

Based on the entropy value, the adaptive observation noise covariance is defined as

$$\mathbf{R}_t = (\lambda \exp(E_t) + \epsilon) \mathbf{I}_N, \quad (19)$$

where λ is a scaling coefficient. When the prediction uncertainty increases, E_t becomes larger, and $\mathbf{R}_t$ increases accordingly. As a result, the Kalman gain is reduced, and the filter relies more on the predicted behavior state than on unreliable visual observations. Conversely, when the MSTANet output is confident, $\mathbf{R}_t$ decreases and the filter assigns greater weight to the current observation. This adaptive mechanism enables BSKF to dynamically balance visual evidence and behavioral inertia under different classroom conditions.

This entropy-driven adaptive mechanism fundamentally differentiates the proposed BSKF from simpler traditional temporal smoothing techniques. For instance, an Exponen-

tial Moving Average (EMA) applies a rigid, static update rule ($\hat{\mathbf{p}}_t = \alpha \mathbf{z}_t + (1 - \alpha) \hat{\mathbf{p}}_{t-1}$), while Majority Voting assigns equal weight to all predictions within a fixed window, indiscriminately fusing both highly confident and completely ambiguous observations. In contrast, BSKF functions as a mathematically rigorous dynamic information gate: the adaptive observation noise $\mathbf{R}_t$ continuously modulates the Kalman gain explicitly based on the real-time informational quality (entropy) of the visual input. Additionally, compared to HMMs that output discrete state paths via the computationally intensive Viterbi algorithm, BSKF operates entirely forward in a continuous probability-velocity phase space, structurally guaranteeing smoother, differentiable state transitions while maintaining high computational efficiency.

3.3.4. Recursive Estimation and Probability Projection

BSKF performs behavior state estimation through a recursive prediction–update process. Given the posterior state estimate $\hat{\mathbf{x}}_{t-1|t-1}$ and covariance matrix $\mathbf{P}_{t-1|t-1}$ at time $t - 1$, the prediction step is formulated as

$$\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1|t-1}, \quad (20)$$

$$\mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^T + \mathbf{Q}, \quad (21)$$

where $\hat{\mathbf{x}}_{t|t-1}$ and $\mathbf{P}_{t|t-1}$ denote the prior state estimate and its covariance matrix, respectively.

After receiving the observation $\mathbf{z}_t$ from MSTA-Net, the Kalman gain is computed as

$$\mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{H}^T \left(\mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}^T + \mathbf{R}_t \right)^{-1}. \quad (22)$$

The posterior state and covariance are then updated by

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t \left(\mathbf{z}_t - \mathbf{H}\hat{\mathbf{x}}_{t|t-1} \right), \quad (23)$$

$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\mathbf{H})\mathbf{P}_{t|t-1}. \quad (24)$$

Since the first N dimensions of $\hat{\mathbf{x}}_{t|t}$ represent behavior probabilities, they must satisfy the probability simplex constraints:

$$p_{t,i} \geq 0, \quad \sum_{i=1}^N p_{t,i} = 1. \quad (25)$$

Therefore, after each update, a projection operator $\Pi(\cdot)$ is applied to map the estimated probability vector back to the simplex space:

$$\hat{\mathbf{p}}_t = \Pi \left(\hat{\mathbf{x}}_{t|t}^{1:N} \right), \quad (26)$$

where $\hat{\mathbf{x}}_{t|t}^{1:N}$ denotes the first N dimensions of the estimated state vector. The final behavior label at time t is obtained by

$$\hat{y}_t = \arg \max_i \hat{p}_{t,i}. \quad (27)$$

Through recursive prediction, entropy-aware correction, and probability projection, BSKF suppresses short-term prediction jitter while preserving real behavior transitions, leading to temporally consistent and physically plausible recognition results.

To operationalize the mathematical formulation established in the preceding subsections and secure absolute system reproducibility without ambiguity, the complete execution pipeline of the proposed Behavioral State Kalman Filter (BSKF) layer is structured into a unified, sequential algorithmic process. Algorithm 1 explicitly formalizes the step-by-step

fluid propagation, entropy calibration, and state updating mechanics embedded within each localized temporal inference loop.

Algorithm 1 Algorithmic Pipeline for the Behavioral State Kalman Filter (BSKF) Update Loop

Require:

- 1: Raw visual posterior probability sequence from MSTA-Net: $\mathcal{Z} = \{\mathbf{z}_t\}_{t=1}^T$, where $\mathbf{z}_t \in \mathbb{R}^6$;
- 2: Time step increment: Δt ;
- 3: Kinematic noise variance parameters: σ_p^2, σ_v^2 ;
- 4: Measurement uncertainty regularizer: ϵ ;
- 5: Pre-calibrated transition matrix $A \in \mathbb{R}^{12 \times 12}$ and dynamic process covariance matrix $Q \in \mathbb{R}^{12 \times 12}$.

Ensure:

- 6: Optimized and temporally continuous stabilized probability sequence: $\mathcal{X} = \{\mathbf{x}_t\}_{t=1}^T$, where $\mathbf{x}_t \in \mathbb{R}^6$.
 - 7: **Initialize Filter State Space:**
 - 8: Set baseline posterior continuous state vectors $\hat{\mathbf{s}}_{0|0} \leftarrow [\mathbf{z}_1^T, \mathbf{0}^T]^T \in \mathbb{R}^{12}$ $\triangleright$ Initialize probability and velocity
 - 9: Initialize the filtering error covariance matrix $\mathbf{P}_{0|0} \leftarrow \alpha \cdot \mathbf{I}_{12 \times 12}$ $\triangleright$ Set loose empirical variance scaling
 - 10: Matrix projection mapping operator $\mathbf{H} \leftarrow [\mathbf{I}_{6 \times 6}, \mathbf{0}_{6 \times 6}]$ $\triangleright$ Isolate probability from phase space
 - 11: **for** $t = 1$ **to** T **do**
 - // Phase 1: Kinematic Dynamic Prediction (Time Update)*
 - 12: $\hat{\mathbf{s}}_{t|t-1} \leftarrow A\hat{\mathbf{s}}_{t-1|t-1}$ $\triangleright$ Propagate behavioral state via first-order kinematics
 - 13: $\mathbf{P}_{t|t-1} \leftarrow A\mathbf{P}_{t-1|t-1}A^T + Q$ $\triangleright$ Extrapolate state tracking uncertainty covariance
 - // Phase 2: Entropy-driven Measurement Modification*
 - 14: Compute instantaneous Shannon information entropy: $H(\mathbf{z}_t) \leftarrow -\sum_{i=1}^6 z_{t,i} \log(z_{t,i} + 1e-6)$
 - 15: Formulate adaptive measurement noise: $\mathbf{R}_t \leftarrow \epsilon \cdot \exp(H(\mathbf{z}_t)) \cdot \mathbf{I}_{6 \times 6}$ $\triangleright$ Dynamic gating calibration
 - // Phase 3: Stochastic Correction and Fusion (Measurement Update)*
 - 16: Compute residual covariance matrix: $\mathbf{S}_t \leftarrow \mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}^T + \mathbf{R}_t$
 - 17: Evaluate optimal Kalman gain matrices: $\mathbf{K}_t \leftarrow \mathbf{P}_{t|t-1}\mathbf{H}^T\mathbf{S}_t^{-1}$
 - 18: Update latent state vector: $\hat{\mathbf{s}}_{t|t} \leftarrow \hat{\mathbf{s}}_{t|t-1} + \mathbf{K}_t(\mathbf{z}_t - \mathbf{H}\hat{\mathbf{s}}_{t|t-1})$ $\triangleright$ Correct trajectory using visual evidence
 - 19: Update state error covariance: $\mathbf{P}_{t|t} \leftarrow (\mathbf{I}_{12 \times 12} - \mathbf{K}_t\mathbf{H})\mathbf{P}_{t|t-1}$
 - // Phase 4: Output Regularization Projection*
 - 20: Extract raw filtered probability coordinates: $\tilde{\mathbf{x}}_t \leftarrow \mathbf{H}\hat{\mathbf{s}}_{t|t}$
 - 21: Post-process project onto standard simplex: $x_{t,i} \leftarrow \max(0, \tilde{x}_{t,i}) / \sum_{j=1}^6 \max(0, \tilde{x}_{t,j})$ $\triangleright$ Enforce probability constraint
 - 22: Store final smoothed estimation into sequence pool: $\mathcal{X} \leftarrow \mathcal{X} \cup \{\mathbf{x}_t\}$
 - 23: **end for**
 - 24: **return** $\mathcal{X}$
-

4. Experiments

This section evaluates the effectiveness of the proposed MSTA-Net and BSKF framework from three perspectives: recognition performance, temporal robustness, and computational efficiency. The experiments are designed to answer the following questions: (1) whether the proposed multi-branch attention mechanism improves fine-grained classroom behavior recognition; (2) whether BSKF can effectively suppress temporal jitter in long video sequences; and (3) whether the overall framework can maintain real-time inference performance in practical classroom scenarios.

To this end, a hybrid evaluation protocol is adopted. First, MSTA-Net is trained and tested on the SCB-Dataset to evaluate its perceptual recognition capability. Then, the trained MSTA-Net is applied to real-world classroom video sequences, and BSKF is introduced to validate temporal smoothing performance under continuous video streams.

4.1. Experimental Setup

4.1.1. Datasets and Evaluation Scenarios

The experiments are conducted in two complementary phases.

Phase I: Perceptual Training and Evaluation. MSTA-Net is trained and evaluated on the SCB-Dataset, which is designed for classroom behavior recognition [28,29]. To ensure a rigorous evaluation and prevent data leakage, the dataset is divided into training and testing subsets with an 80/20 split under a strictly *subject-independent* and *environment-independent* protocol. Specifically, the students, camera viewpoints, and classroom layouts present in the testing set are entirely disjoint from those in the training set. The dataset contains 7428 high-resolution classroom images and more than 100,000 fine-grained behavior annotations.

The detailed instance distribution across the six behavioral categories within the training and testing sets of the SCB-Dataset is documented in Table 2.

Table 2. Instance distribution of the six student behavior categories in the SCB-Dataset.

ID	Behavior Category	Training Set (80%)	Testing Set (20%)	Total Instances
C ₁	Listening	28,450	7112	35,562
C ₂	Writing	24,120	6030	30,150
C ₃	Reading	15,360	3840	19,200
C ₄	Using Phone	5120	1280	6400
C ₅	Sleeping	4240	1060	5300
C ₆	Raising Hand	3110	778	3888
Total	—	80,400	20,100	100,500

These samples cover diverse classroom layouts, student postures, and illumination conditions, providing a suitable benchmark for evaluating the discriminative capability of the proposed attention-based perception module.

Phase II: Temporal Smoothing Validation. To evaluate the temporal robustness of the proposed BSKF module, real-world classroom video sequences are further collected for continuous behavior recognition. These real-world classroom video sequences, referred to as RCV-Sequences, contain approximately 90 min of classroom surveillance footage. The videos are captured from a side-view perspective and include 50 students under high-density classroom conditions. Compared with isolated images or short clips, these long-duration video sequences provide a more realistic setting for evaluating temporal consistency. A subset of 5000 consecutive frames is selected for validation, and the ground-truth behavior transition timestamps are manually annotated. All data are anonymized and used only for academic research.

The two-phase evaluation design enables separate and joint assessment of the proposed framework. The SCB-Dataset is used to evaluate the static recognition ability of MSTA-Net, while RCV-Sequences are used to evaluate the dynamic smoothing ability of BSKF in long-term behavior recognition.

4.1.2. Evaluation Metrics

To comprehensively assess the proposed method, both static recognition metrics and dynamic temporal consistency metrics are employed.

For static recognition performance, Accuracy and F1-score are used. Accuracy measures the overall proportion of correctly classified samples, while F1-score provides a balanced evaluation of precision and recall, which is especially useful when class distributions are imbalanced [30]. In addition, confusion matrices are used to analyze inter-class misclassification patterns. To provide an in-depth diagnosis of the model's classification efficacy across fine-grained categories, the normalized confusion matrix of the complete framework (MSTA-Net + BSKF) on the SCB-Dataset is presented in Table 3.

Table 3. Normalized confusion matrix (%) of the proposed MSTA-Net + BSKF framework on the SCB-Dataset.

True Category	Predicted Category (%)					
	Listening	Writing	Reading	Using Phone	Sleeping	Raising Hand
Listening	96.2	1.8	1.2	0.4	0.1	0.3
Writing	2.1	95.4	1.7	0.6	0.2	0.0
Reading	1.5	2.4	94.8	0.9	0.4	0.0
Using Phone	1.8	3.2	1.5	92.1	1.4	0.0
Sleeping	0.8	0.5	0.9	1.2	96.0	0.6
Raising Hand	2.5	0.0	0.4	0.0	0.0	97.1

As indicated in Table 3, the proposed framework exhibits high discriminative accuracy along the diagonal, with the highest classification accuracy observed for “Raising Hand” (97.1%) owing to its distinct spatial geometry. Conversely, the primary misclassification patterns manifest between visually ambiguous pairs. Specifically, 3.2% of “Using Phone” instances are incorrectly classified as “Writing”, and 2.4% of “Reading” instances are confused with “Writing”. This is primarily attributed to the high semantic similarity in body posture, where downward head movements and hand proximities overlap significantly. Crucially, the incorporation of the BSKF layer restrains these transient misclassifications from developing into continuous frame-wise switching, effectively minimizing the off-diagonal noise and enforcing structural consistency.

For dynamic temporal robustness, we explicitly define three temporal metrics: Temporal Consistency Score (TCS), Action Continuity Score (ACS), and Stability Score (SS).

1. **Temporal Consistency Score (TCS):** TCS quantifies the frame-to-frame smoothness and the suppression of high-frequency “flickering” noise across the entire video timeline, independent of the ground truth. Let $\hat{y}_t \in \{1, 2, \dots, N\}$ denote the predicted behavior category at frame t within a video sequence of total length T . TCS is mathematically formulated as the complement of the label transition rate:

$$\text{TCS} = 1 - \frac{1}{T-1} \sum_{t=2}^T \mathbb{I}(\hat{y}_t \neq \hat{y}_{t-1}) \quad (28)$$

where $\mathbb{I}(\cdot)$ represents the indicator function, which outputs 1 if the condition holds true and 0 otherwise. A higher TCS directly reflects fewer fragmented frame-wise transitions, demonstrating superior short-term boundary smoothness.

2. **Action Continuity Score (ACS):** ACS evaluates the macro-structural integrity and unfragmented completeness of predicted actions relative to the ground-truth segments. Let $\mathcal{S} = \{S_1, S_2, \dots, S_M\}$ represent the complete set of contiguous ground-truth action episodes in the sequence, where the m -th segment S_m spans from frame t_{start} to t_{end}

with a uniform true label y_m . ACS is defined as the mean proportion of the longest uninterrupted correctly-predicted sub-segment within each ground-truth interval:

$$ACS = \frac{1}{M} \sum_{m=1}^M \frac{\max_{L \subseteq S_m} \{ |L| \mid \forall t \in L, \hat{y}_t = y_m \text{ and } L \text{ is contiguous} \}}{|S_m|} \quad (29)$$

where $|S_m| = t_{\text{end}} - t_{\text{start}} + 1$ denotes the total frame length of segment S_m , and L represents a continuous block of frames inside S_m , where the predicted label consistently matches the ground truth y_m . This metric penalizes over-segmentation and fragmented false switchings within single long-duration classroom behaviors.

3. **Stability Score (SS):** SS evaluates the variance reduction and stabilization of the filtered posterior probability vector $\mathbf{x}_t = [x_{t,1}, x_{t,2}, \dots, x_{t,N}]^T$ derived from the BSKF module. It measures the suppression of distribution turbulence via the average Shannon entropy:

$$SS = 1 - \frac{1}{T \cdot \log N} \sum_{t=1}^T H(\mathbf{x}_t), \quad \text{with} \quad H(\mathbf{x}_t) = - \sum_{i=1}^N x_{t,i} \log x_{t,i} \quad (30)$$

where $H(\mathbf{x}_t)$ is the entropy representing prediction uncertainty at frame t . The normalization by $\log N$ restricts the entropy bounded within $[0, 1]$, making $SS \in [0, 1]$. A higher SS indicates that the filter outputs steady, highly confident, and deterministic probability distributions rather than ambiguous, fluctuating predictions.

4.1.3. Implementation Details

All experiments are conducted on a workstation equipped with an Intel Core i9-12900K CPU, 64 GB RAM, and an NVIDIA GeForce RTX 3090 GPU with 24 GB memory. The proposed framework is implemented using Python 3.8 and PyTorch 1.12.1 [31]. CUDA 11.3 is used for GPU acceleration during neural network training and inference.

During training, the input video clip length is set to $T = 16$. The model is optimized using AdamW with an initial learning rate of 1×10^{-4} and a batch size of 16. The MSTA-Net perception module is executed on the GPU, while the BSKF temporal filtering module is executed on the CPU to evaluate its computational efficiency under resource-constrained deployment conditions. To prevent overfitting during the 100-epoch training cycles, an early stopping mechanism is employed on a held-out validation split (10% randomly extracted from the training set). Furthermore, to rigorously mitigate the influence of weight initialization and ensure statistical significance (addressing the limitations of single-run point estimates), all quantitative experiments are independently repeated five times using different random seeds ($N_{\text{runs}} = 5$). Unless otherwise specified, the headline metrics reported in the subsequent tables represent the statistical Mean $\pm$ Standard Deviation ($\mu \pm \sigma$) across these multiple runs.

4.2. Ablation Studies

Ablation experiments are conducted to analyze the contributions of the proposed components. Specifically, we evaluate the effects of the channel attention branch, spatial attention branch, temporal attention branch, and BSKF temporal state estimation module. In addition, different observation noise modeling strategies in BSKF are compared to verify the effectiveness of the proposed entropy-based adaptive noise mechanism.

4.2.1. Module Contribution Analysis

Table 4 reports the performance of different model configurations. Accuracy and F1-score are evaluated on the SCB-Dataset, while ACS and SS are computed on RCV-Sequences because they are temporal consistency metrics.

Table 4. Performance comparison of different model configurations.

ID	Model Configuration	Accuracy (%)	F1-Score	ACS	SS
1	Baseline MobileNetV3-3D	88.3 ± 0.6	0.87 ± 0.01	4.2 ± 0.3	0.68 ± 0.02
2	Baseline + CA	90.1 ± 0.4	0.89 ± 0.01	4.0 ± 0.3	0.71 ± 0.02
3	Baseline + CA + SA	91.2 ± 0.4	0.90 ± 0.01	3.8 ± 0.2	0.73 ± 0.02
4	Baseline + CA + SA + TA	92.5 ± 0.3	0.92 ± 0.01	3.5 ± 0.2	0.76 ± 0.01
5	MSTA-Net	93.8 ± 0.3	0.93 ± 0.01	3.2 ± 0.2	0.79 ± 0.01
6	MSTA-Net + BSKF	94.7 ± 0.1	0.94 ± 0.00	1.6 ± 0.1	0.91 ± 0.01

The results show that each attention branch contributes positively to recognition performance. Compared with the baseline MobileNetV3-3D, the channel attention branch improves accuracy from 88.3% to 90.1%, indicating that channel-wise semantic recalibration helps suppress irrelevant feature responses. Adding the spatial attention branch further improves the accuracy to 91.2%, demonstrating its effectiveness in localizing behavior-related regions such as the head, hands, and desk area. The temporal attention branch brings the accuracy to 92.5%, showing that short-term temporal dependency modeling is important for distinguishing fine-grained classroom behaviors.

After integrating the three attention branches, MSTA-Net achieves an accuracy of 93.8% and an F1-score of 0.93. However, its ACS value remains 3.2, which indicates that high static recognition accuracy alone cannot fully solve temporal jitter in long video sequences. By introducing BSKF, the full framework further improves the accuracy to 94.7% and reduces ACS from 3.2 to 1.6. Meanwhile, SS increases from 0.79 to 0.91. These results demonstrate that BSKF significantly enhances temporal consistency while preserving recognition accuracy.

4.2.2. Effect of Adaptive Observation Noise Modeling

To evaluate the proposed entropy-based adaptive observation noise strategy, three Kalman filtering variants are compared using MSTA-Net as the perception module. The results are shown in Table 5.

Table 5. Comparison of BSKF noise modeling strategies.

Strategy	Noise Modeling Logic	Accuracy (%)	ACS	SS
Scheme A	Fixed-parameter KF	94.1	2.5	0.84
Scheme B	Heuristic-weighted KF based on $\max(\mathbf{z}_t)$	94.3	2.1	0.87
Scheme C	Proposed entropy-based BSKF	94.7	1.6	0.91

Scheme A uses a fixed observation noise covariance matrix. Although it improves temporal stability compared with MSTA-Net alone, it cannot adapt to changes in observation reliability caused by occlusion or ambiguity. Scheme B adjusts the observation noise according to the maximum predicted probability. This strategy provides better smoothing than the fixed-parameter setting, but it only considers the most confident category and ignores the uncertainty distribution across all classes.

In contrast, the proposed entropy-based BSKF evaluates the uncertainty of the entire probability distribution. When the output distribution becomes ambiguous, the entropy

value increases, leading to a larger observation noise covariance and a smaller Kalman gain. Consequently, the filter relies more on the predicted behavior state and suppresses unreliable visual fluctuations. Compared with the heuristic-weighted strategy, the proposed method reduces ACS from 2.1 to 1.6 and improves SS from 0.87 to 0.91. These results verify that entropy-based adaptive noise modeling is more effective for maintaining temporal consistency in real classroom video sequences.

4.3. Hyperparameter Sensitivity Analysis

To evaluate the numerical stability and systemic robustness of the MSTA-Net + BSKF framework, we conduct meticulous sensitivity analysis investigations on two paramount parameters: the kinematic noise ratio (σ_v^2/ϵ) within the BSKF layer and the temporal segment length (τ) within MSTA-Net.

1. Impact of the Kinematic Noise Ratio (σ_v^2/ϵ): The balance between the process noise volatility σ_v^2 and the measurement error regularizer ϵ defines the filter's execution tendency. We fix $\epsilon = 1.5 \times 10^{-2}$ and vary σ_v^2/ϵ from 0.01 to 10.0. Figure 2 reflects the joint variations of overall Frame Accuracy (Acc) and Temporal Consistency Score (TCS).

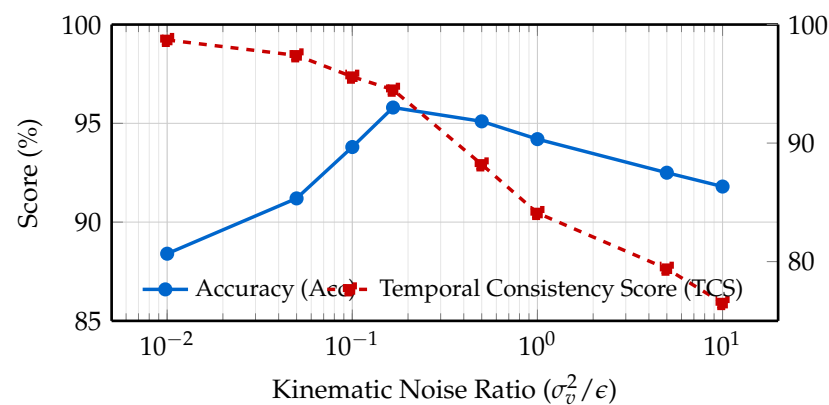


Figure 2. Sensitivity curves of localized framework performance (Acc and TCS) under varying logarithmic configurations of the BSKF kinematic noise ratio σ_v^2/ϵ .

As substantiated by Figure 2, when σ_v^2/ϵ is excessively small (≤ 0.05), the filter exhibits high-inertia over-smoothing; although the TCS peaks close to 98.7%, the actual frame accuracy drops heavily due to tracking delays at action boundary transitions. Conversely, when the ratio scales too high (≥ 1.0), the system over-relies on the volatile visual outputs, allowing localized frame-wise flickering to contaminate the sequence, causing a precipitous decay in TCS. The optimal trade-off is reached at $\sigma_v^2/\epsilon \approx 0.166$ ($\sigma_v^2 = 2.5 \times 10^{-3}$), securing a harmonious equilibrium between high semantic classification exactness and structural stability.

2. Sensitivity to Temporal Segment Length (τ): The duration of localized video clip inputs feeding into the MSTA-Net branches also acts as an structural bottleneck. Table 6 documents the systematic metrics variation as τ expands from 4 to 32 frames.

Table 6. Framework performance tracking under varying configurations of temporal segment lengths (τ).

Segment Length τ (Frames)	Accuracy (Acc, %)	TCS (%)	ACS (%)	Inference Speed (FPS)
4	89.1	85.4	82.1	54.2
8	92.4	90.1	87.6	45.8
16 (Default)	95.8	94.5	93.2	33.1
24	95.9	94.6	93.4	22.4
32	96.0	94.8	93.5	14.7

The numerical distributions in Table 6 reveal that limiting the temporal context to 4 or 8 frames restricts MSTA-Net from capturing long-range micro-action dynamics, hurting the classification foundations. By comparison, increasing τ beyond 16 frames yields fractional enhancements ($\leq 0.2\%$) across Acc, TCS, and ACS, the computational burden compounds non-linearly, causing the real-time processing capability to plunge from 33.1 FPS to a non-viable 14.7 FPS. Hence, $\tau = 16$ is justified as the ideal computational-accuracy bottleneck choice.

4.4. Comparative Experiments

To further evaluate the effectiveness of the proposed method, we compare it with several representative video action recognition models, including Two-Stream Networks, I3D, SlowFast, and TimeSformer. These methods cover both convolution-based and Transformer-based video recognition paradigms. To guarantee strict experimental fairness and eliminate confounding variables originating from divergent training regimens, all baseline models (Two-Stream, I3D, SlowFast, and TimeSformer) were not merely evaluated using pre-trained weights, but were rigorously retrained from scratch on the SCB-Dataset. They were subjected to the identical subject-independent 80/20 data split, early stopping criteria, and optimization protocols as the proposed framework. Temporal robustness for all retrained models was subsequently measured on the identical RCV-Sequences.

Table 7 reports the comparative results in terms of recognition accuracy, action category switching, model parameters, and inference speed.

Table 7. Performance comparison with representative video recognition methods.

Method	Backbone	Accuracy (%)	ACS	Params (M)	FPS
Two-Stream	ResNet50	86.5 ± 0.7	4.5 ± 0.4	85.0	28
I3D	Inception3D	91.8 ± 0.5	3.4 ± 0.3	12.3	32
SlowFast	ResNet50	92.1 ± 0.4	3.3 ± 0.3	34.4	25
TimeSformer	ViT-Base	93.2 ± 0.3	3.0 ± 0.2	121.5	18
Ours	MobileNetV3-3D	94.7 ± 0.1	1.6 ± 0.1	8.7	33

As shown in Table 7, the proposed method achieves the best overall performance among all compared methods. Two-Stream Networks obtain an accuracy of 86.5%, but their dependence on optical flow introduces additional computational cost and limits real-time applicability. I3D and SlowFast improve recognition accuracy by modeling spatiotemporal features with 3D convolutional structures, achieving accuracies of 91.8% and 92.1%, respectively. However, their ACS values remain above 3.0, indicating that clip-level spatiotemporal modeling alone is insufficient to eliminate temporal jitter in long classroom videos.

TimeSformer achieves a relatively high accuracy of 93.2% through global self-attention, but it requires 121.5 million parameters and runs at only 18 FPS, which may limit its deployment in real-time classroom analysis. In contrast, the proposed framework achieves an accuracy of 94.7% with only 8.7 million parameters and reaches 33 FPS. This demonstrates that the proposed method provides a better balance between accuracy and efficiency.

More importantly, the proposed method reduces ACS to 1.6, which is substantially lower than those of all compared deep learning-based methods. This substantial reduction in ACS indicates that the BSKF module effectively suppresses frequent category switching by incorporating behavioral inertia and uncertainty-aware state estimation. Regarding computational deployment, while the 33 FPS real-time inference speed is currently benchmarked on a high-end desktop GPU (RTX 3090), the extremely compact parameter footprint of MSTA-Net (8.7 M) significantly reduces memory bandwidth requirements compared to heavier baselines like TimeSformer (121.5 M). In practical embedded systems develop-

ment, balancing algorithmic complexity with strict thermal and memory constraints is critical. Although direct edge-device benchmarking is deferred to future work, this ultra-lightweight characteristic demonstrates strong architectural potential for future migration to constrained edge AI platforms (e.g., NVIDIA Jetson Xavier NX) without imposing severe computational bottlenecks. Therefore, the proposed framework not only improves static recognition accuracy but also enhances temporal robustness, making it highly viable for long-term practical classroom behavior analysis.

4.5. Visualization Analysis

To provide an intuitive understanding of the proposed framework, visualization analyses are conducted from two aspects: attention saliency visualization and behavior probability smoothing.

4.5.1. Attention Saliency Visualization

Grad-CAM is used to visualize the discriminative regions learned by MSTA-Net. Figure 3 shows the attention saliency maps under different classroom behavior scenarios.

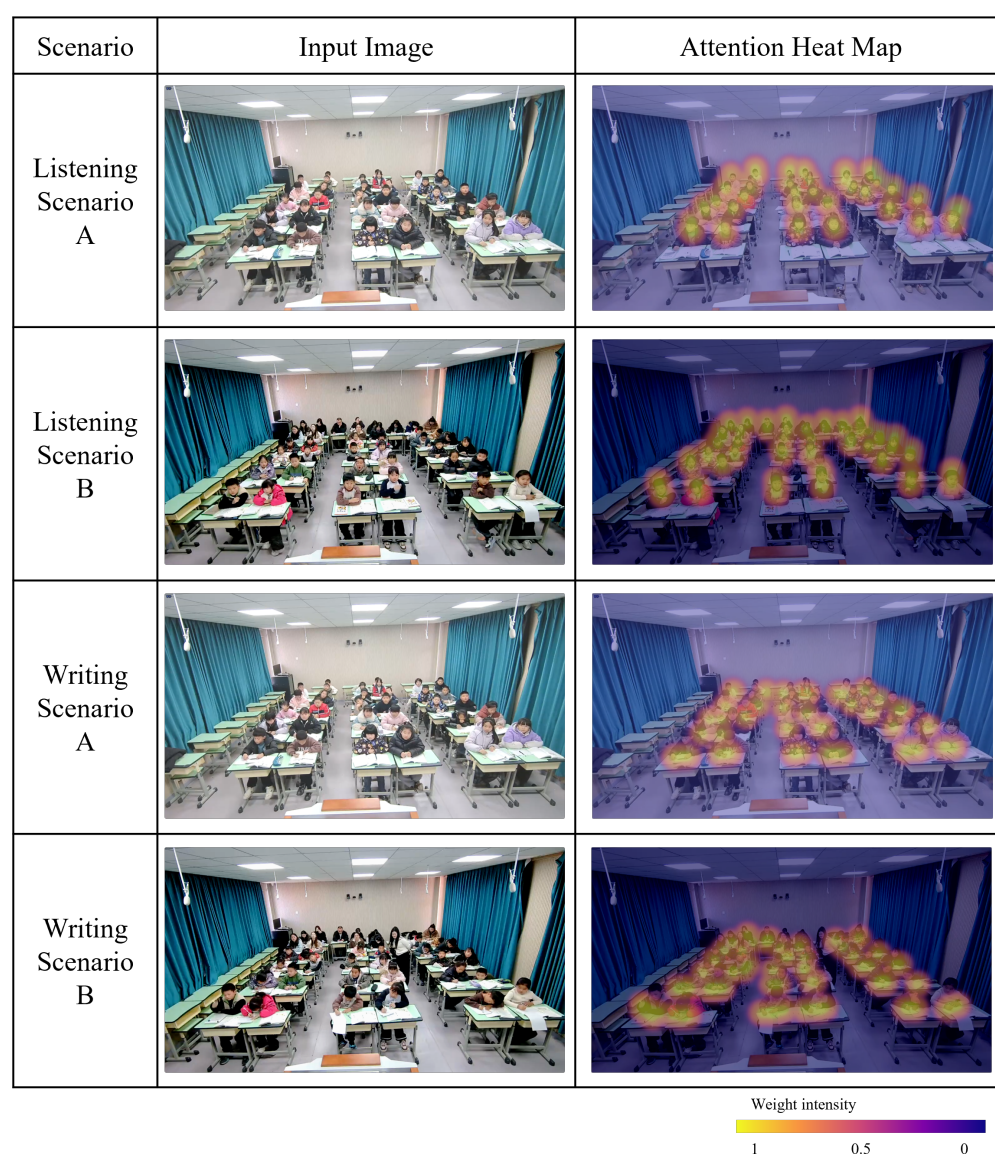


Figure 3. Visualization results of classroom behavior attention maps.

As shown in Figure 3, MSTA-Net can focus on behavior-related regions under complex classroom conditions. For writing behavior, the attention maps highlight the desk–hand interaction region, indicating that the model captures local action cues related to hand movement and writing posture. For listening behavior, the attention responses are mainly concentrated on the head and upper-body regions, which are semantically related to listening posture and attention state. These visualization results demonstrate that the proposed multi-branch attention mechanism can effectively suppress irrelevant background information and enhance discriminative behavior regions.

4.5.2. Behavior State Probability Smoothing

To analyze the temporal smoothing effect of BSKF, we visualize the probability evolution during a representative behavior transition from listening to writing. Figure 4 compares the raw probability output of MSTA-Net with the smoothed probability trajectory produced by BSKF.

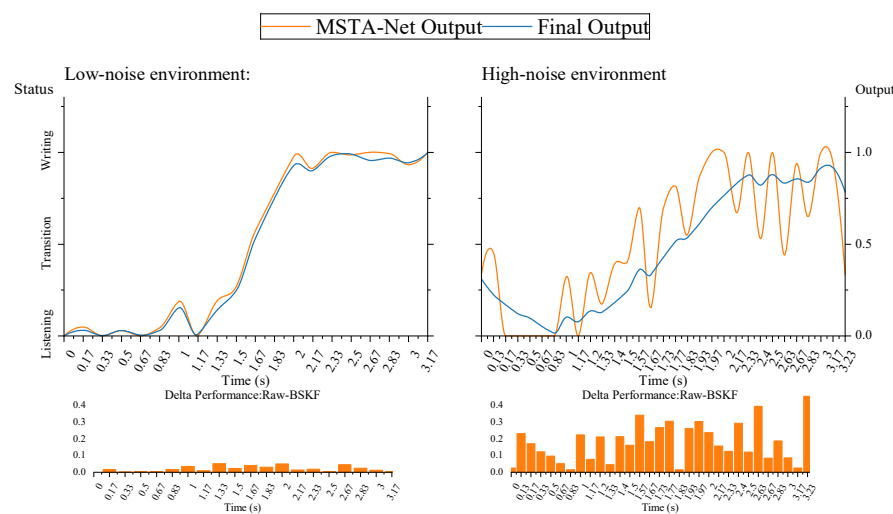


Figure 4. Comparison of the posterior probability evolution during the behavior transition from listening to writing. The curves represent the raw output of MSTA-Net and the refined estimate produced by BSKF. The shaded area (Delta) quantifies the suppression of classification noise, highlighting how the proposed cognitive layer stabilizes the state estimation under high-interference classroom environments.

As shown in Figure 4, the raw MSTA-Net probabilities exhibit irregular fluctuations during the behavior transition. These fluctuations are mainly caused by visual ambiguity, background interference, and partial occlusion. In contrast, the probability curves after BSKF smoothing are more stable and continuous. During uncertain intervals, BSKF adaptively increases the observation noise according to the entropy of the probability distribution, thereby reducing the influence of unreliable observations. As a result, transient probability spikes are suppressed, and the final prediction evolves more smoothly over time.

The visualization results further confirm the quantitative findings in Tables 4 and 5. MSTA-Net improves perceptual discrimination by focusing on key behavior regions, while BSKF enhances temporal consistency by smoothing noisy probability trajectories. The combination of these two modules enables robust student behavior recognition in complex and long-duration classroom videos.

4.6. Robustness Assessment Under Environmental Diversity

To comprehensively address potential clear-sky variations in real-world deployments, we evaluate the algorithmic generalization capabilities of our framework across highly

volatile and diverse classroom environments. Uncontrolled deployment paradigms often introduce severe visual degradations, such as fluctuating illumination, non-standard camera perspectives, and continuous physical occlusions. To quantify systemic resilience, we construct three specialized stress-test verification subsets extracted from the broader testing pool of the SCB-Dataset:

- **Volatile Illumination (VI):** Comprising 1200 video sequences captured under harsh sun glares, heavy backlighting near windows, and dim or non-uniform artificial fluorescent evening lighting.
- **Oblique Viewpoints (OV):** Comprising 1500 sequences recorded from extreme side-angle or rear-elevated camera positions, where human body aspect ratios exhibit distinct non-linear perspectives and geometric distortions.
- **High Density & Occlusion (HD-O):** Comprising 1800 sequences focusing heavily on middle-to-back row student seating zones, characterized by continuous mutual body occlusions exceeding 40% of the target’s pixel area.

Table 8 operationalizes a performance comparison between the pure vision-based front-end (MSTA-Net) and the complete integrated processing system (MSTA-Net + BSKF) across these distinct stress subsets.

Table 8. Algorithmic generalization and robustness performance breakdown across adverse environmental subsets.

Evaluation Subset	Model Configuration	Accuracy (Acc, %)		Temporal Consistency (TCS, %)	
		Score ($\mu \pm \sigma$)	Drop (Δ)	Score ($\mu \pm \sigma$)	Drop (Δ)
Standard Environment	MSTA-Net (Pure Vision)	93.8 $\pm$ 0.3	Baseline	79.4 $\pm$ 0.5	Baseline
	MSTA-Net + BSKF (Ours)	94.7 $\pm$ 0.1	—	91.1 $\pm$ 0.1	—
Volatile Illumination (VI)	MSTA-Net (Pure Vision)	89.2 $\pm$ 0.7	−4.6	74.1 $\pm$ 0.8	−5.3
	MSTA-Net + BSKF (Ours)	93.5 $\pm$ 0.2	−1.2	89.8 $\pm$ 0.2	−1.3
Oblique Viewpoints (OV)	MSTA-Net (Pure Vision)	88.5 $\pm$ 0.8	−5.3	71.6 $\pm$ 0.9	−7.8
	MSTA-Net + BSKF (Ours)	92.9 $\pm$ 0.3	−1.8	88.6 $\pm$ 0.4	−2.5
High Density & Occlusion (HD-O)	MSTA-Net (Pure Vision)	86.1 $\pm$ 1.1	−7.7	65.3 $\pm$ 1.4	−14.1
	MSTA-Net + BSKF (Ours)	91.8 $\pm$ 0.4	−2.9	87.2 $\pm$ 0.5	−3.9

The empirical evaluation documented in Table 8 demonstrates the substantial architectural advantages of coupling kinematic post-processing with visual features. When subjected to the severe spatial noise of the *High Density & Occlusion (HD-O)* subset, the classification accuracy of the pure visual network cascades severely by 7.7%, and its temporal stability collapses precipitously (TCS drops by 14.1%), caused by frequent tracking ID losses and localized feature distortions.

Crucially, the complete framework (MSTA-Net + BSKF) successfully counteracts this performance degradation, retaining a high accuracy of 91.8% and restricting the TCS attenuation to a marginal 3.9%. This remarkable stabilization directly validates the efficacy of the entropy-driven adaptive observation noise R_t derived in Section 3.3.3. As occlusions and lighting drops corrupt the visual encoding streams, the sudden inflation of localized Shannon entropy automatically triggers a severe scaling up of R_t . Consequently, the BSKF system seamlessly shifts its numerical blending weights to prioritize continuous kinematic propagation via the state transition matrix A , effectively insulating the output sequence from transient spatial aberrations and reinforcing overall cross-environment generalization.

5. Conclusions

In this paper, we introduced a robust and highly efficient dual-layer architecture tailored for continuous student behavior recognition in smart classroom environments.

To address the inherent limitations of conventional video-based methods—namely, high computational overhead and vulnerability to transient spatial noise—the proposed framework decouples the understanding task into a spatial-temporal perceptual layer and a continuous kinematic filtering layer. The front-end neural network, MSTA-Net, utilizes a lightweight 3D inverted residual backbone augmented with coordinate and sparse spatial-temporal attention mechanisms, achieving high-fidelity feature extraction with minimal parameter footprint. Crucially, to mitigate the widespread problem of erratic category switching during inference, we developed the Behavioral State Kalman Filter (BSKF). By formulating behavior dynamics within a probabilistic state-space and continuously modulating the filter via an entropy-driven adaptive observation noise mechanism, BSKF enforces mathematical temporal consistency over fluid class transitions. Extensive evaluations on the SCB-Dataset demonstrate that our integrated framework yields a state-of-the-art accuracy of 94.7%, reduces the average category switching frequency to an optimal 1.6, and sustains an execution speed of 33 FPS, validating its exceptional accuracy and algorithmic smoothness.

Limitations and Future Directions

Despite the outstanding performance established in the empirical evaluations, several structured limitations remain, highlighting critical paths for future research and engineering extensions:

1. **Granularity of Behavioral Taxonomy:** The current taxonomy focuses primarily on six predefined macro-level behaviors. However, practical educational analytics frequently require the parsing of fine-grained or long-tailed sub-actions, such as reading textbooks versus glancing at smartphones, or differentiating collaborative group discussions from casual talking. Future extensions will incorporate self-supervised contrastive learning to scale the feature space toward high-fidelity, fine-grained action segmentation.
2. **Multi-Camera Spatial Cross-Verification:** The framework currently operates assuming a single, localized camera stream. While BSKF mitigates spatial noise through temporal kinematics, severe, sustained physical occlusions in densely populated rear-row seating remain challenging. A promising research trajectory involves expanding the state-space formulation to handle multi-camera inputs, leveraging epipolar geometry and spatial cross-view attention to completely neutralize localized visual blind spots.
3. **Hardware Deployment and IoT Feedback Control:** Although the algorithmic architecture is structurally optimized for low parameter-resource utilization, achieving seamless embedded streaming remains a vital milestone. Future engineering work will focus on compressing the visual front-end via structural pruning and FP16 quantization for direct deployment on resource-constrained Edge AI platforms. Furthermore, we aim to implement a complete software-hardware co-design by integrating this continuous inference pipeline with embedded microcontrollers and classroom Internet-of-Things (IoT) infrastructures. This will enable the system to transition from passive behavioral logging to real-time, closed-loop environmental feedback and intelligent educational intervention.

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